TERRAPROOF | SATELLITE EVIDENCE DOSSIER

Autonomous Geospatial Audit & Decision-Support System • Generated 2026-11-01 05:24:44 UTC

CRYPTOGRAPHIC INTEGRITY HASH (SHA-256): 8919daa0d1f9105a1b5ffbace8cd09b552b06dd0fb90bbaf7d0103345fdaf084

Audit Dossier ID:	test-job-uuid-12345	**Analysis Type:**	INSURANCE VERIFICATION
Claim Reference:	TEST-REF-9921	**State / District:**	Punjab / Ludhiana
Farmer / Insured:	Sunita Devi	**Crop & Season:**	Soybean (Kharif)
Khasra / Land Record:	N/A	**Sum Insured / ha:**	■35,000/ha
Recommended Verdict:	**MANUAL_FIELD_INSPECTION_RECOMMENDED**	**Evaluation Standard:**	PMFBY Geospatial Audit Standard

1. EXECUTIVE EVIDENCE FINDINGS

Satellite analysis observed 440.236 ha (100.0% of field) exhibiting spectral signatures consistent with flood damage (440.24 ha severe loss, 0.0 ha moderate stress).

Metric	Value	Methodological Meaning
Baseline Mean Index	0.8839	Mean radiometric canopy baseline prior to reported event
Post-Event Mean Index	-0.3386	Mean radiometric response observed immediately following event
Index Difference (Delta)	1.2225	Magnitude of spectral divergence (positive = loss/inundation)
Estimated Affected Area	**440.236 ha**	Total hectares showing thresholded anomalous change
Claimed Loss Area	8.4 ha	Hectares reported in submission form
Confidence Score	**91%**	HIGH_CONFIDENCE (91/100): Reliable satellite evidence layer; fast-track desk adjustment eligible.

2. METEOROLOGICAL CONTEXT (OPEN-METEO ARCHIVE)

Ground coordinates (27.11, 76.21) recorded **68.4 mm** total rainfall during the target window, peaking at 41.2 mm/day. Average topsoil moisture was estimated at 0.38 m³/m³.

3. SATELLITE SENSORS & ACQUISITIONS USED

Scene ID	Sensor	Acquisition Date	Cloud Cover
DEMO_S2A_MSIL2A_20260710_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-10T10:30:00Z	4.2%
DEMO_S2B_MSIL2A_20260710_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-10T10:35:00Z	8.5%
DEMO_S2A_MSIL2A_20260715_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-15T10:30:00Z	4.2%
DEMO_S2B_MSIL2A_20260715_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-15T10:35:00Z	8.5%

4. POTENTIAL INCONSISTENCIES & AUDIT FLAGS

• ***AUDIT FLAG:*** *Observed affected area (440.236 ha) diverges by 5140.9% from reported claimed area (8.4 ha).*

5. METHODOLOGICAL LIMITATIONS & RESPONSIBLE USE

• Civilian satellite pixel resolution is 10 meters; perimeter boundary mix may introduce +/- 8% area uncertainty.
• Surface greenness (NDVI) can decrease due to seasonal crop harvesting rather than natural disaster.
• Weather archive data represents grid centroid rainfall; localized cloudbursts may show slight microclimate divergence.

6. ESTIMATED COMPENSATION SCENARIO (PMFBY MODEL)

Sum Insured / ha	■35,000	**Assumed Severity %**	80.0%
Observed Affected Area	440.24 ha	**Estimated Scenario Amount**	**■12,326,608**

Formula: ■35,000/ha × 440.24 ha × 0.80 = ■12,326,608 (Range: ■11,340,479 – ■13,312,737) • Payout range reflects satellite pixel resolution and confidence uncertainty. Final PMFBY claim settlement requires survey committee ratification.

7. HUMAN ADJUSTER AUDIT & DECISION SIGN-OFF

Adjuster Decision:	PENDING ACCREDITED REVIEW	**Date of Inspection:**	_____
Adjuster Name & ID:	_____ __	**Physical Signature:**	_____

LEGAL & FORENSIC DISCLAIMER: *TerraProof is a technical decision-support and evidence-gathering instrument. Satellite spectral indices (such as NDVI, NDWI) provide indirect radiometric indicators of vegetation and surface moisture. This dossier does NOT constitute a legally binding verdict of fraud or guarantee of compensation. Final claim approval or denial remains the exclusive responsibility of accredited human insurance adjusters.*